

15 D $\lambda = 2(0.90) = 1.80 \text{ m}$

$$\frac{\Delta\phi}{2\pi} = \frac{\Delta x}{\lambda}$$

$$\Delta\phi = (360^\circ) \left(\frac{\Delta x}{\lambda} \right)$$

$$= (360^\circ) \left(\frac{1.30}{1.80} \right) 360$$

$$= 260^\circ$$

16 D check for max observable order for violet: